Si5351A VFO Rev A — Field Manual Ch. 6. Plan: PLAN.md (2026-10-22). Hierarchy: power/USB, MCU/UI, synth/RF.
Inter-sheet nets via global labels (MCU_TX, MCU_RX, DTR, I2C_SCL, I2C_SDA) and power rails.

Power and USB

File: power_usb.kicad_sch

MCU and UI

File: mcu_ui.kicad_sch

Synth and RF outputs

File: synth_rf.kicad_sch

Draft schematic, not reviewed. See VERIFY notes.

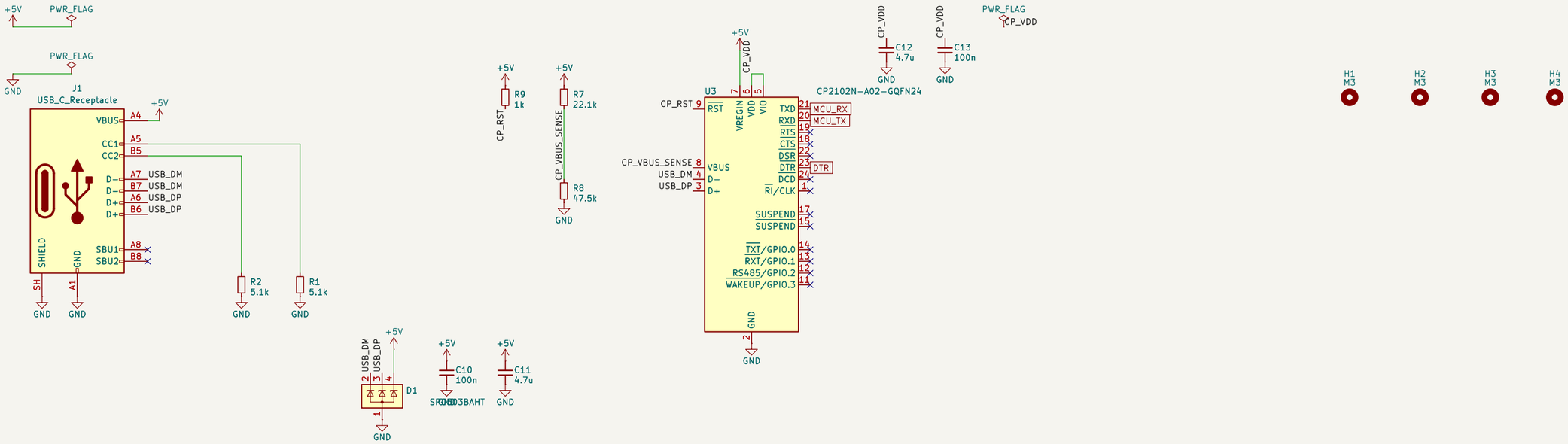
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Sheet: /
File: Si5351AVFO.kicad_sch

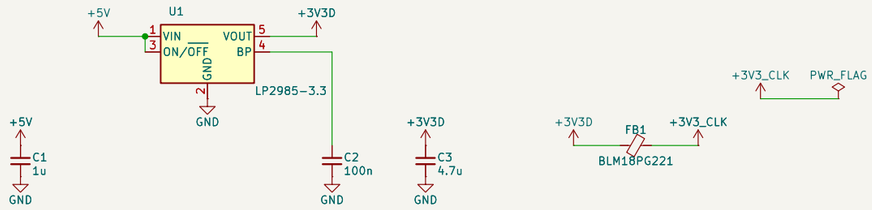
Title: Si5351A VFO — Top

Size: A3	Date: 2026-10-22	**Rev: A**
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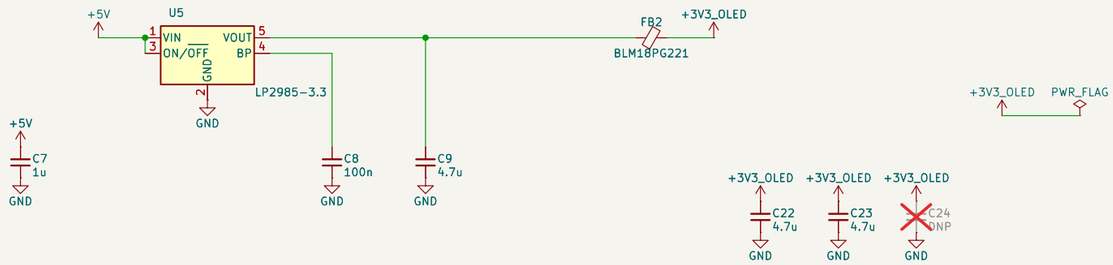
USB-C input, CP2102N USB-UART (from EtherkitUARTAdapter), LP2985 rails (U1 from Rev F; U5 dedicated to OLED per Jason 2026-10-22)



U1: +3V3D (MCU) and, via FB1, +3V3_CLK (Si5351 + TCXO). Rev F U1/C1/C2/C3.



U5: OLED-only rail (Eyes On The Prize lesson, PLAN 4a). Place U5/FB2/C22-C24 at the OLED header.



VERIFY (unconfirmed items on this sheet):
- J1: UART adapter schematic pairs GCT USB4105 footprint with this HCTL LCSC part; HCTL footprint used here — confirm. Shield to GND assumed (not stated in PLAN).
- D1: Pin-to-line mapping (D-/D+/VBUS on pins 2/3/4) to be checked against adapter layout.
- C10: LCSC stock low; fallback C49678 (0805) per PLAN
- U3: Unused CP2102N pins left NC (adapter used DTR via JP1, TRRS dropped). Confirm none were strapped on the adapter.
- R9: PLAN says /RST pull-up to VBUS (as on adapter); CP2102N RST max is VIO-referenced — confirm adapter really ties to VBUS.
- FB2: Bead or series R (<=6 ohm upper bound, PLAN 4a); value to be set by measurement.
- C22: OLED bulk value: PLAN start point 2x4.7u, tune by measurement
- C24: DNP 1206 site for larger OLED bulk cap; value/part not chosen

Draft schematic, not reviewed. See VERIFY notes.

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Sheet: /Power and USB/
File: power_usb.kicad_sch

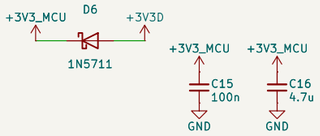
Title: Si5351A VFO — Power and USB

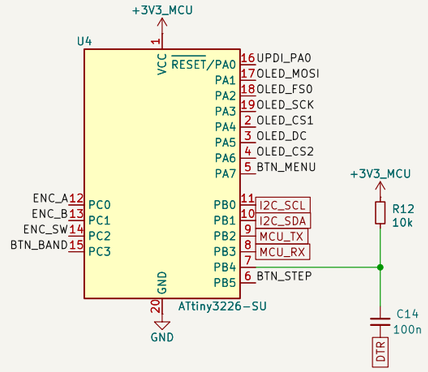
Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/4

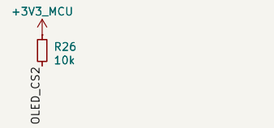
ATtiny3226 @ 8 MHz internal, 3.3 V. Optiboot over CP2102N; DTR auto-reset on PB4 (RSTPINCFG=0x3, megaTinyCore "UPDI w/reset on PB4"), as EtherKeyerMini C3/R4.



+3V3_MCU = +3V3D via D6 OR UPDI pin 1 via D5 (ATtiny only).

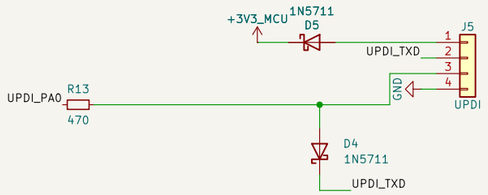


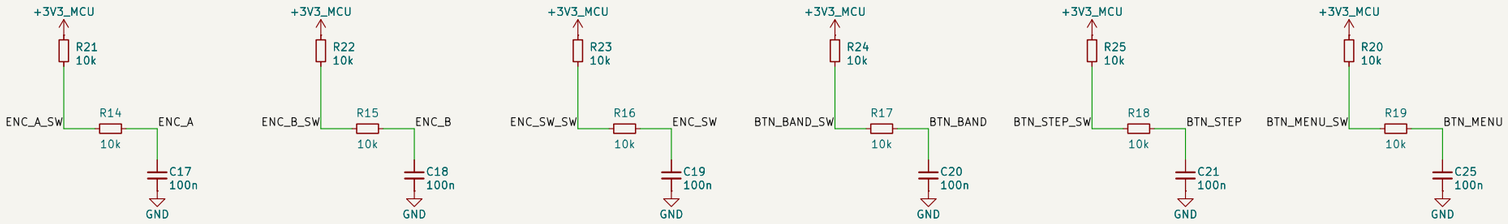


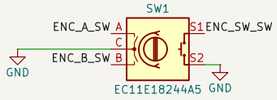


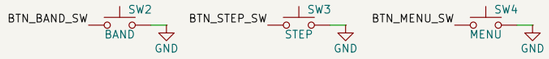
OLED: SPI only (PA1 MOSI, PA3 SCK, PA4 CS1, PA5 DC). Never shares the Si5351 I2C bus.
CS2 (font chip) held high by R26.

UPDI / serial-UPDI header, circuit copied from EtherKeyerMini (J3,R3,D1,D2 -> J5,R13,D4,D5;
EKM JP2/JP3 omitted per Jason). Pin 1 VCC (via D5, feeds +3V3_MCU only), 2 TXD, 3 RXD/UPDI, 4 GND.









VERIFY (unconfirmed items on this sheet):
- U4: Pin mux (PLAN sec.3) to be checked vs datasheet I/O mux table. Jason consigning parts.
- D6: ATtiny-only rail +3V3_MCU: diode-OR of +3V3D (D6) and UPDI header VCC (D5) so a programmer cannot back-feed +3V3D. Same 1N5711 as EKM D5 (no LCSC/MPN). MCU VDD -2.9-3.0 V (0.3-0.4 V drop); check ATtiny VIH/VIL vs CP2102N TXD (3.3 V VIO, exceeds MCU VDD by ~0.35 V) and OLED/SI5351 3.3 V logic; I/O pull-ups moved to +3V3_MCU. With only programmer power, MCU pins can back-power unpowered CP2102N/OLED/SI5351 via their ESD diodes.
- C14: PLAN: series R or clamp on PB4 for DTR-rising overshoot? EKM has none.
- J5: Replicated from EtherKeyerMini (J3): EKM has no LCSC/MPN fields - part number to choose.
- D5: Replicated from EtherKeyerMini (D2): EKM has no LCSC/MPN fields - part number to choose.
- R13: Replicated from EtherKeyerMini (R3): EKM has no LCSC/MPN fields - part number to choose.
- D4: Replicated from EtherKeyerMini (D1): EKM has no LCSC/MPN fields - part number to choose.
- DS1: 1.3in SPI SH1106 module. Pin order GND,VCC,SCL,SDA,DC,CS1,FS0,CS2 from PLAN pin list; pin numbering, pitch and connector style unverified. Low stock - Jason sourcing.
- C17: Debounce topology/count: PLAN lists x5 but 6 inputs (A,B,SW,3 buttons) - drawn as 6. Check vs encoder bounce spec.
- SW1: Detent/pulse count unverified.

Draft schematic, not reviewed. See VERIFY notes.

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Sheet: /MCU and UI/
File: mcu_ui.kicad_sch

Title: Si5351A VFO - MCU and UI

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/4

Si5351A + 25 MHz TCXO copied from Si5351A Breakout Rev F (R5/C5/C6/R6/C4, XB open). Each output: Rs 0R -> Cb 100n DC block -> DNP shunt -> LPF (DNP) with 0R bypass -> SMA.

VERIFY (unconfirmed items on this sheet):
- X1: TCXO stock thin (718): buy before ordering.
- R27: I2C pull-up value not fixed (PLAN 2.2-4.7k); no LCSC chosen. Pulled to +3V3_MCU (not +3V3_CLK) so SCL/SDA never exceed Attiny VDD; SI5351 VIH 0.7*VDD=2.31 V vs ~2.95 V high - check.
- R28: as R27
- R100: 0603 0R series (Jason decision); LCSC code not in PLAN (C17477 is 0805).
- C100: DC-block LF corner ~32 kHz into 50 ohm (PLAN Q6).
- R101: DNP shunt ~120 ohm back-match of ~85 ohm source (datasheet Zo); part/LCSC not chosen; verify return loss esp. >100 MHz.
- L1: LPF inductor size (SMD vs T37 option, PLAN Q7)
- C105: Parallel C across L (elliptic) per LowPassFilterModule - unconfirmed
- C101: LPF values builder choice (DNP)
- J2: Footprint is a placeholder; check BWSMA-KE-P001 launch geometry and 1.6 mm edge fit.
- R120: see R100
- R121: see R100
- J3: see J2
- R140: see R100
- R141: see R100
- J4: see J2

+3V3_CLK
R5
10
PWR_FLAG
R6
0
C5
100n
GND
X1
25MHz TCXO
NC/EN
VDD
GND
OUT
C6
100n
+3V3_CLK
C4
100n
GND
+3V3_CLK
U2
XA
XB
VDD
VDDO
CLK0
CLK1
CLK2
GND
I2C_SCL
I2C_SDA
SCL
SDA
Si5351A-B-GT
+3V3_MCU
R27
VERIFY 2.2k-4.7k
I2C_SCL
+3V3_MCU
R28
VERIFY 2.2k-4.7k
I2C_SDA

Bypass: RC (R104/R124/R144) fitted by default; RA/RB DNP. Filter build: fit RA/RB, remove RC.
Keep RA/RB pads <=2 mm from junction (PLAN 6).

CLK0 output chain (J2)

CLK0
R100
0
C100
100n
R101
120
GND
R102
0
C101
DNP
C105
DNP
L1
DNP
C102
DNP
C106
DNP
L2
DNP
C103
DNP
C107
DNP
L3
DNP
C104
DNP
R103
0
J2
SMA CLK0
R104
0

CLK1 output chain (J3)

CLK1
R120
0
C120
100n
R121
120
GND
R122
0
C121
DNP
C125
DNP
L4
DNP
C122
DNP
C126
DNP
L5
DNP
C123
DNP
C127
DNP
L6
DNP
C124
DNP
R123
0
J3
SMA CLK1
R124
0

CLK2 output chain (J4)

CLK2
R140
0
C140
100n
R141
120
GND
R142
0
C141
DNP
C145
DNP
L7
DNP
C142
DNP
C146
DNP
L8
DNP
C143
DNP
C147
DNP
L9
DNP
C144
DNP
R143
0
J4
SMA CLK2
R144
0

Draft schematic, not reviewed. See VERIFY notes.

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Sheet: /Synth and RF outputs/
File: synth_rf.kicad_sch

Title: Si5351A VFO - Synth and RF outputs

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/4